Project Plan: Real-Time IoT Data Pipeline

1. Project Summary

Item	Detail
Project	Real-Time IoT Data Pipeline
Duration	4 weeks
Team Size	5 members
Team Leader	Rana Shaaban Abdelqader Mohamed
Repository	https://github.com/nhahub/NHA-5-053.git

2. Timeline Overview

Week	Phase	Goal
1	Data Simulation and Ingestion	Sensor data is generated and flows into Kafka
2	Batch Data Pipeline (ETL)	Data is cleaned, transformed, and stored in Snowflake
3	Streaming Pipeline with Alerts	Real-time threshold alerts are working
4	Dashboard and Final Report	Live dashboard, final report, and presentation are ready

3. Gantt Chart

■ = work period for that task. Each week is split into two halves (A and B).

Task	Owner	W1-A	W1-B	W2-A	W2-B	W3-A	W3-B	W4-A	W4-B
Define data schema	Mostafa	■							

Task	Owner	W1-A	W1-B	W2-A	W2-B	W3-A	W3-B	W4-A	W4-B
Data simulator (every 5 sec)	Mostafa	■	■						
Flask API	Waleed	■	■						
Kafka setup and producer	Waleed	■	■						
Environment setup (Kafka, Snowflake, Python)	All	■	■						
Kafka consumer for batch	Abdelrahman		■	■					
Batched files output	Abdelrahman		■	■					
ETL: extract and transform	Shahd			■	■				
ETL: load into Snowflake	Shahd			■	■				
Data validation and optimization	Shahd				■				
Streaming consumer and threshold rules	Mostafa					■	■		
Alert logic and testing with anomalies	Mostafa					■	■		
Pipeline performance monitoring	Abdelrahman						■		
Dashboard build	Rana				■	■	■	■	
Connect dashboard to processed and streamed data	Rana						■	■	
Integration testing (end to end)	All						■	■	
Final report	All (Rana compiles)							■	■
Presentation and rehearsal	All								■

4. Milestones

#	Milestone	Target	Done When
M1	Data generator and schema ready	End of Week 1	Simulator produces valid readings every 5 seconds and they reach Kafka
M2	Batch ETL complete	End of Week 2	Transformed data is loaded into Snowflake and validated
M3	Streaming alerts working	End of Week 3	Threshold breaches trigger alerts during the anomaly test
M4	Dashboard live	Middle of Week 4	Dashboard shows sensor trends, anomaly counts, and alert history
M5	Final delivery	End of Week 4	Report, slides, and architecture summary are on GitHub and the demo is rehearsed

5. Deliverables per Week

Week 1

- Python generator script
- Sample data logs
- Data schema documentation
- Environment setup notes

Week 2

- ETL script
- Processed dataset in storage
- Data validation report
- ETL documentation

Week 3

- Streaming pipeline setup
- Alert logic code and output
- Real-time query configurations

- Streaming architecture documentation

Week 4

- Dashboard screenshot or live demo
- Final project report
- Presentation slides
- System architecture summary

6. Resource Allocation

Member	Week 1	Week 2	Week 3	Week 4
Rana (Team Leader)	Coordination, environment setup	Coordination, dashboard design	Dashboard development	Dashboard, final report, presentation
Abdelrahman	Kafka consumer (start)	Kafka consumer, batched files	Performance monitoring	Testing, report
Mostafa	Data simulator, schema	Support testing	Streaming consumer and alerts	Testing, report
Shahd	Snowflake setup	ETL to Snowflake	Validation fixes	Testing, report
Waleed	Flask API, Kafka producer	Producer fixes and support	Support streaming tests	Testing, report

Tools and infrastructure: Python, Pandas, Flask, Apache Kafka, Snowflake, Streamlit (or Power BI / Grafana), Git and GitHub.

7. Communication and Tracking

- Short weekly planning meeting at the start of each week and a review at the end.
- Quick progress check-ins between members whose parts depend on each other.
- Tasks tracked on a shared board, and all code and documents pushed to GitHub.

- Each member updates their own folder and documentation before the end of the week.

8. Dependencies

1. The data schema must be agreed before the simulator, API, and ETL are built.
2. Kafka must be running before the producer and consumers can be tested.
3. The batch consumer must produce files before the ETL can load them.
4. The streaming consumer needs a stable Kafka topic from the producer.
5. The dashboard needs processed data in Snowflake and the alert stream.